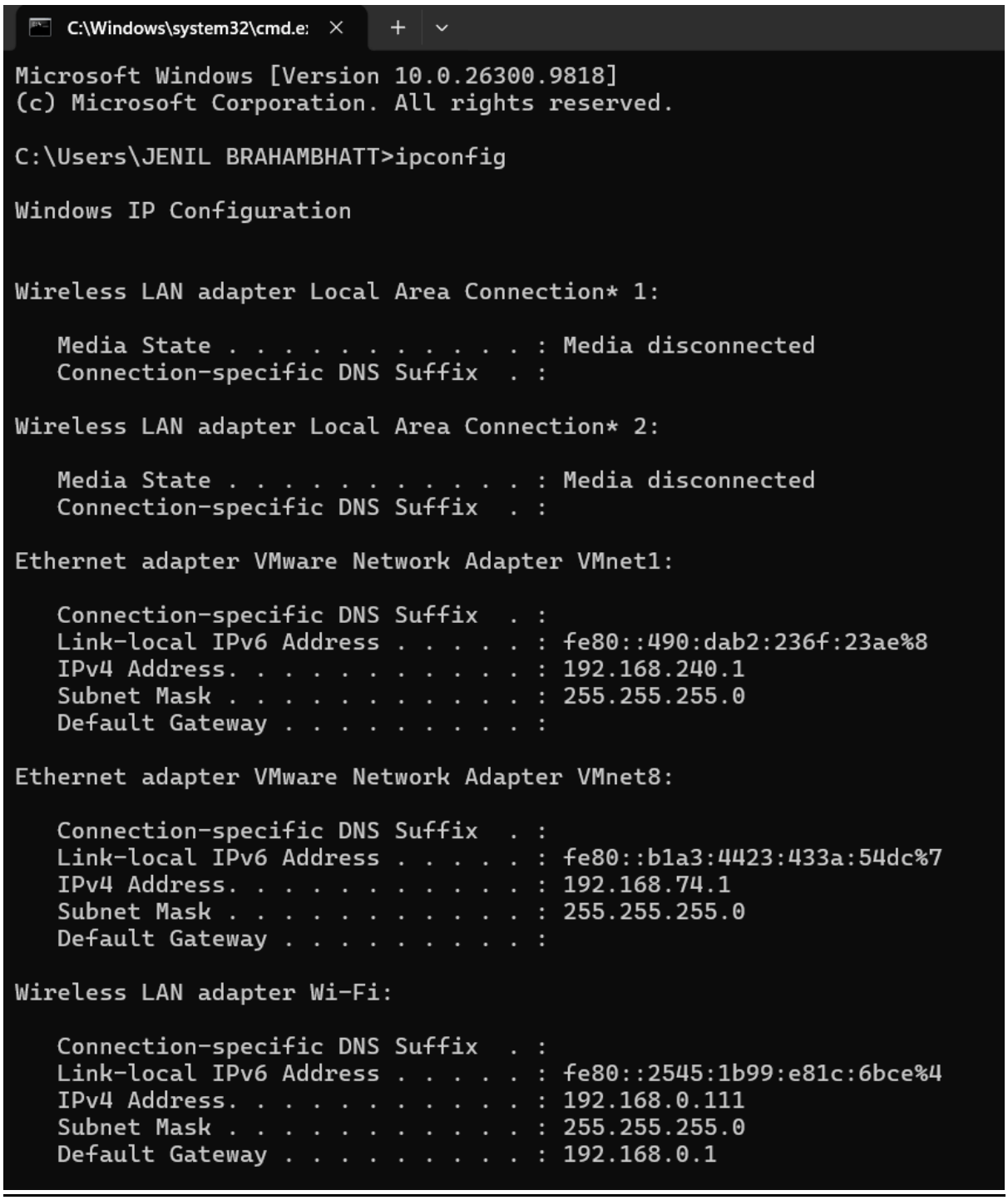
1.

Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

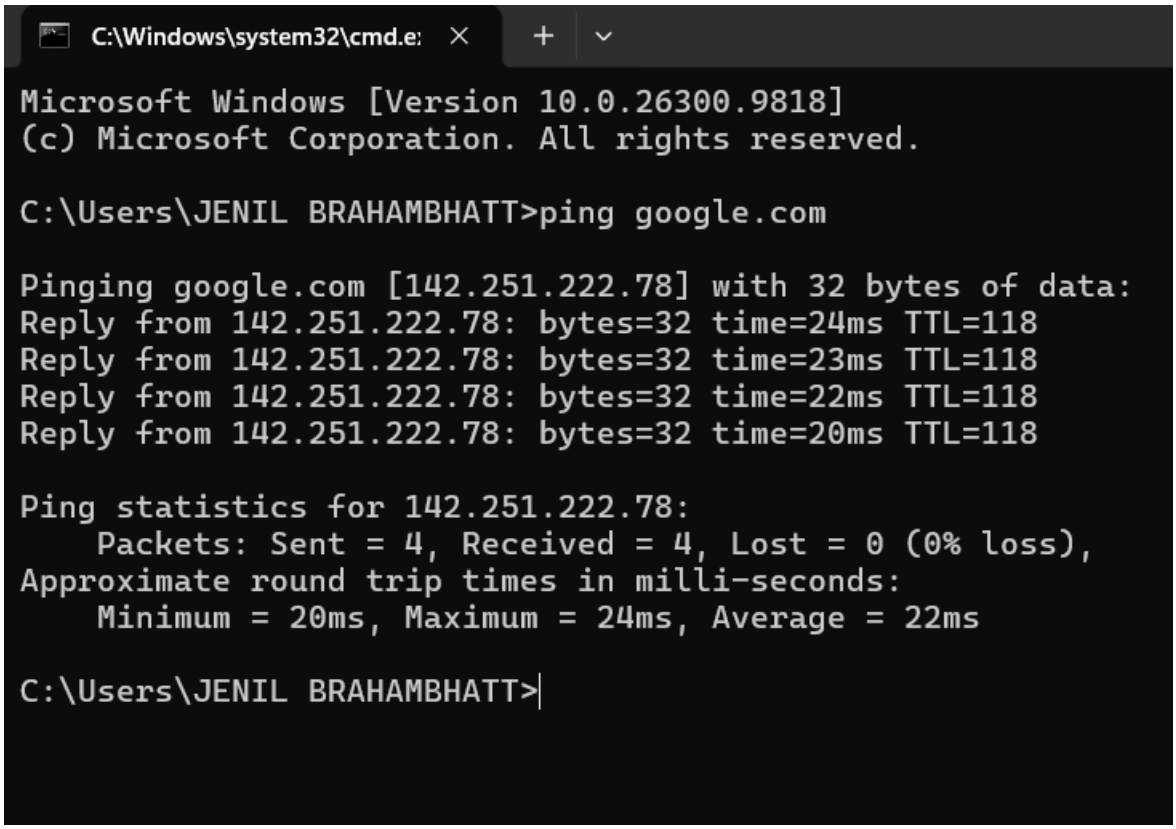
ANS:-



2.

Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

ANS:-



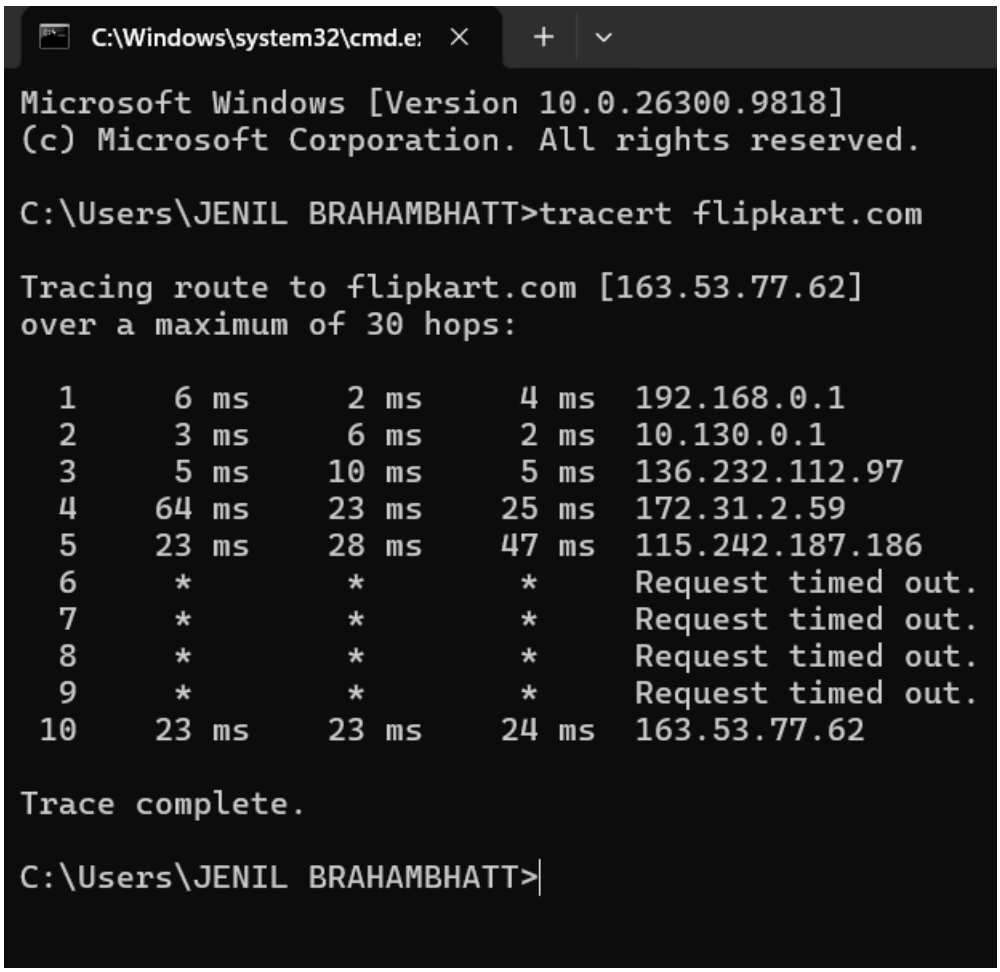
Approximate round trip times in milli-seconds:

Minimum = 20ms, Maximum = 24ms, Average = 22ms

3.

Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

ANS:-

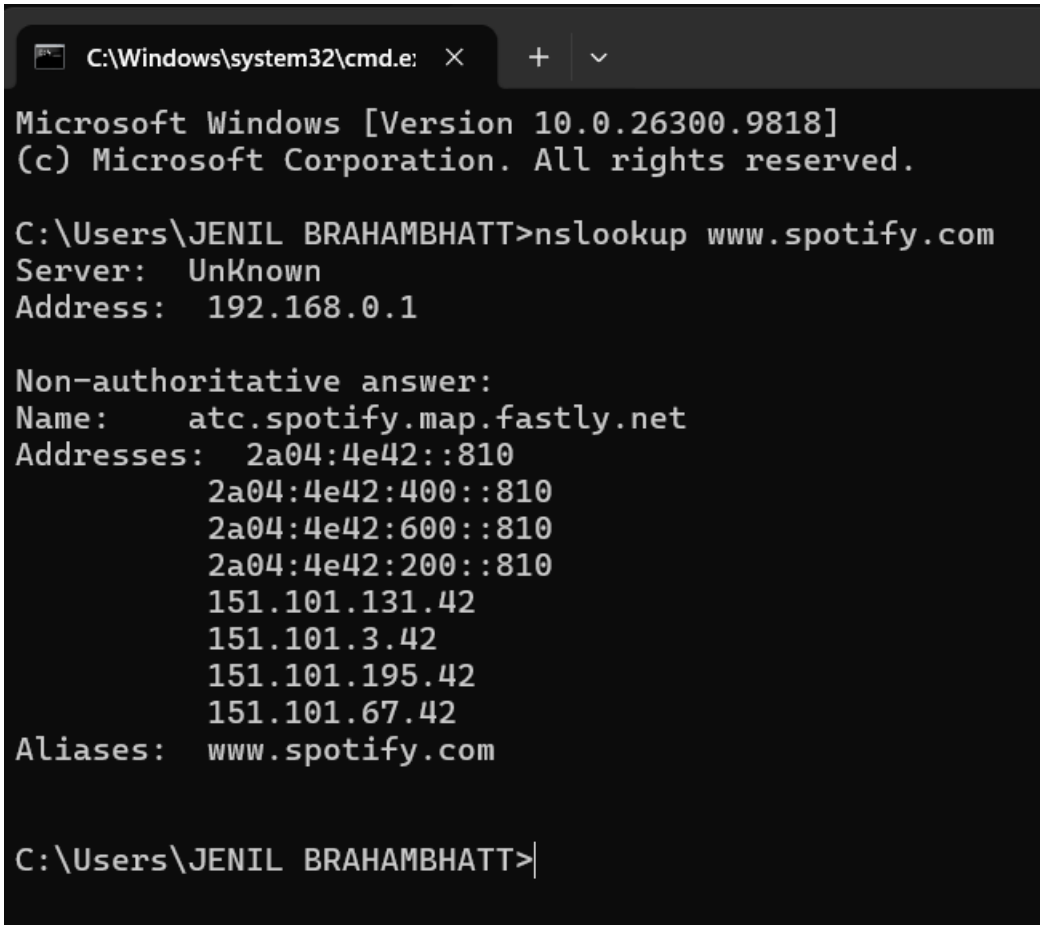


it takes 10 hops to reach Flipkart's destination server (163.53.77.62).

4.

Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

ANS:-



What it helps verify: nslookup helps you verify whether your configured DNS server is successfully translating human-readable domain names into their corresponding IP addresses.

5.

A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check. Hint: Think about checking both network connectivity and DNS resolution.

ANS:-

Our friend cannot access Zomato's website, we can run the following sequence of three diagnostic commands:

1. **ipconfig**

    - **What it checks:** Verifies the computer's local IP configuration, subnet mask, and default gateway, ensuring the device has successfully received an IP address and isn't stuck with an APIPA address (169.x.x.x) due to a DHCP failure.
2. **nslookup www.zomato.com**
    - **What it checks:** Tests DNS resolution to determine if the local DNS server can successfully resolve Zomato's domain name into a valid IP address.
3. **ping www.zomato.com (or ping <IP_address>)**
    - **What it checks:** Tests active network connectivity and measures packet loss and round-trip response time between the laptop and the target server.